

Enhancing Privacy-Preserving Cancer Classification with Convolutional Neural Networks

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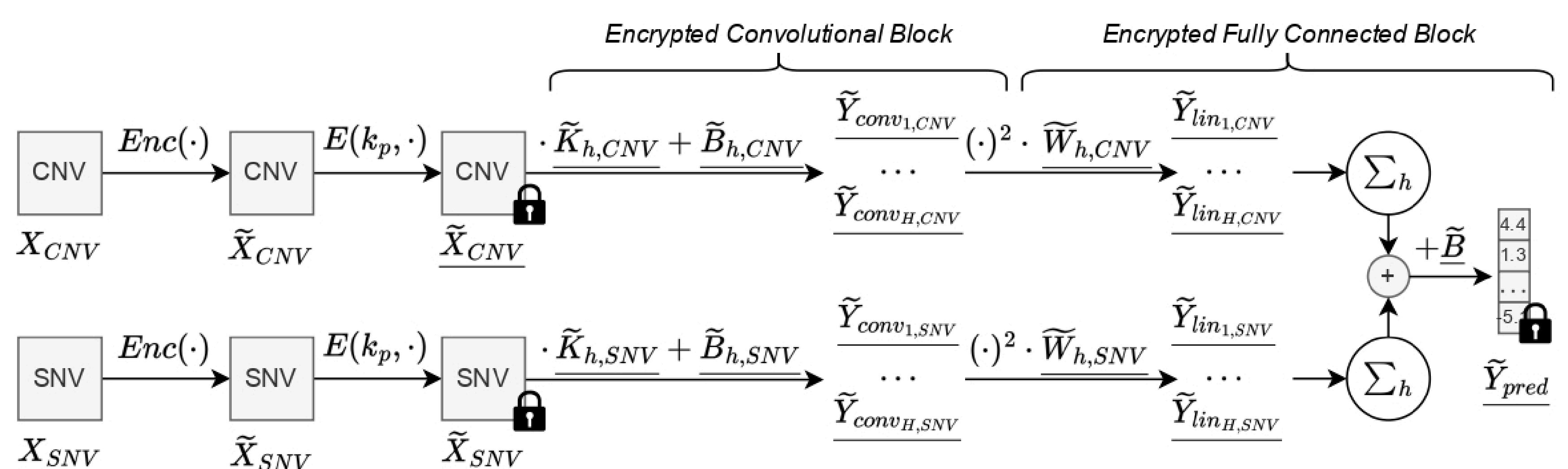
TL;DR

The paper presents OGHE, a privacy-preserving CNN for cancer classification, leveraging homomorphic encryption. It achieves state-of-the-art accuracy and reduced latency on genomic data, ensuring secure AI for precision medicine.

Code



Architecture



Introduction

- OGHE is a privacy-preserving CNN that incorporates parallel one-dimensional convolutional layers to independently capture SNVs and CNVs spatial patterns.
- Our feature selection technique, VarScout, uses clustering and mutation frequency to identify key SNVs and CNVs.
- A novel packing mechanism enables to efficiently encrypt data, weights, and biases into ciphertexts, for high computational performance and reduced latency.

Background

- HE is a cryptographic technique allowing computations to be performed directly on encrypted data without decrypting it, preserving privacy while ensuring secure processing.
- SNVs are genetic alterations involving a single nucleotide, which can act as biomarkers for diseases and influence protein structure and function.
- CNVs are structural genetic variations where segments of DNA are duplicated or deleted, impacting gene expression.

Future Works

- Minimization of the overhead introduced by HE for improved computational efficiency.
- Development of a framework for OGHE encrypted training.
- Exploiting NAS to optimize OGHE's architecture under HE constraints.

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Results

